

**Q2-02-A**

Let

$$f(x) = \sqrt{x}.$$

1. Find the Taylor polynomial of degree 3 for  $f(x)$  about the point  $x_0 = 1$ .
2. We use this Taylor polynomial to approximate

$$f(0.9) \quad \text{and} \quad f(1.1).$$

Use Taylor's theorem with remainder to find an error formula for the Taylor polynomial. Give an upper bound for the error in each of the two approximations.